

The C-E-A Framework for AI Agents: Compliance, Efficacy, and Autonomy — A Case Illustration in Personalized Robotic Care

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Abstract

AI agents increasingly perceive, reason, and act with reduced human oversight, yet accurate reasoning alone says nothing about whether a specific candidate action is fit to deploy: whether it is legally and practically usable, whether it will actually help given real-world constraints, or how much independent action the agent should take before deferring to a person. This paper introduces the C-E-A framework (Compliance, Efficacy, Autonomy), which evaluates one candidate agent action at one decision point against three questions: is the action legally, medically, and practically permitted by everyone who holds standing authority over it (Compliance); will the action actually help right now, given real conditions, once a binary feasibility floor is cleared (Efficacy); and can the agent act independently while the person retains genuine, supported choice, bounded by a documented escalation ceiling and a hard red line against non-consensual physical action (Autonomy). This paper illustrates the framework using personalized robotic care for older adults as a case scenario, and reports the framework's own open tensions, including an unresolved conflict between a caregiver's standing permission and a capable person's stated preference.

Keywords: *C-E-A Framework, AI Agent Evaluation, Compliance, Efficacy, Autonomy, Personalized Robotics, Human-AI Agency, AI Governance, Public Health AI*

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1. Introduction

AI agents are increasingly asked not just to inform but to act — recommending, scheduling, dispatching, and, in embodied settings, physically assisting. As agents take on more of the reasoning and the acting themselves, the central evaluation question shifts: not whether an agent's output is accurate, but whether a specific recommended action is fit to deploy — usable under the law, actually helpful under real conditions, and appropriately bounded in how much the agent does on its own.

Evaluating whether an agent's reasoning is accurate does not answer whether its output is legally and practically usable, whether it actually protects health under real-world constraints, or how much independent action it should take before deferring to a human. Closing this gap requires an

explicit evaluation layer applied to the candidate action itself, not just to the agent's perception or reasoning.

Why Public Health Needs This Beyond the Classic Agent Loop

Public health raises this gap in an especially acute form. Its actions are rarely just correct or incorrect: they are taken on behalf of, or physically upon, people who may be minors, cognitively impaired, uninsured, or otherwise dependent on caregivers, guardians, or institutions with their own standing authority. An agent that reasons correctly but recommends an action outside its legal scope, that cannot be carried out given a person's real housing or mobility constraints, or that oversteps how much independent action a vulnerable person has consented to, produces a public health harm even when its underlying reasoning was accurate — and as agents move from advisory output toward direct, sometimes physical action, these failures stop being hypothetical. This is the significance of C-E-A: an explicit, action-level check for legal and caregiver permission, real-world feasibility, and preserved human agency, applied at exactly the point where the classic agent loop is silent.

This also points to a narrower, more concrete role than broad ethical theory typically occupies — closer to what sociology calls a middle-range framework¹ — which may be the level of precision agentic AI training eventually needs. Whether C-E-A can support that role, rather than only evaluating agent actions after deployment, is left as an open question for future work.

To address this gap, this paper introduces the C-E-A framework (Compliance, Efficacy, Autonomy), positioned later in this paper against bioethical principlism², AI governance vocabularies³, and implementation-science constructs⁴, each of which addresses an adjacent but distinct question.

This paper presents C-E-A's three evaluation pillars together with three supporting mechanisms that anchor them to real-world deployment: an explicit Unit of Analysis for scoring ongoing relationships one action at a time, a hard Feasibility Floor under Efficacy, and a Multi-party Permission layer under Compliance. Applied to personalized robotic care for older adults, these mechanisms show how C-E-A separates legal permission, practical usability, caregiver authority, and preserved human choice into three explicit, jointly necessary evaluation pillars.

2. Conceptual Foundation

The classic AI agent loop — perceive, reason, act — explains how an agent processes an environment.⁵ C-E-A is best understood as an evaluation layer applied on top of this loop, not a replacement for it. The loop describes what an agent mechanically does; C-E-A asks whether what it does is fit for a public-health or personal-care context. Perceiving the environment has no C-E-A layer of its own — it is raw sensing. Reasoning about what should be done is evaluated by Efficacy: will the candidate action actually help, given real constraints? Acting — including physically — is evaluated jointly by Compliance (is this legal, medically appropriate, and permitted?) and Autonomy (how much can the agent do without waiting on a human?).

A second design choice sits underneath all three pillars: C-E-A scores a single candidate action at a single decision point, not an entire agent, an entire relationship, or an entire day. In a one-off event, such as an evacuation alert, the decision point is obvious. In an ongoing personalized relationship, such as a home robot running continuously, the framework applies at each moment the agent considers a specific action — offering physical support, sending an alert, withholding one, escalating to a caregiver. This Unit of Analysis does not require inventing artificial event boundaries: every C-E-A score in this framework is a score of one action-instance, and a single agent accumulates many separate scores over time rather than one lifetime rating.

3. The C-E-A Framework

3.1 Definition and Scope

C-E-A evaluates a candidate action against three jointly necessary pillars (Figure 1). The pillar metaphor is deliberate: like the legs of a table, all three must bear weight for the action to stand. A strong score on one pillar does not compensate for a failing score on another — removing any single pillar leaves the action unable to support deployment, regardless of how solid the remaining two are.

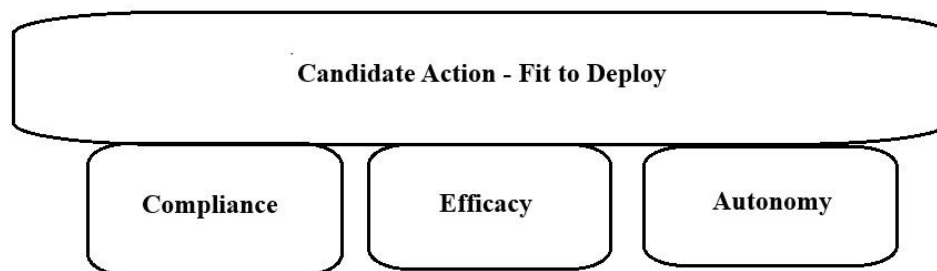


Figure 1. The C-E-A framework as three load-bearing pillars supporting one deployable action

- **Compliance (C):** is this action legally and medically permitted, consistent with the person's own physical constraints, and accepted by everyone who holds standing permission over it? Three sub-components must all be satisfied — legal/regulatory compliance, practical/adoptive compliance, and multi-party permission compliance — or the action is “compliant on paper only.”
- **Efficacy (E):** will this action actually prevent harm or protect health, right now, given real conditions? A binary Feasibility Floor is applied first — can the action physically and practically be carried out in the person's actual space right now? If not, Efficacy is zero regardless of theoretical benefit.
- **Autonomy (A):** can the agent act on its own safely, while the person retains real choice and dignity — and does it ever treat the person as a passive object of its action? A full mark on preserved human agency requires a concrete mechanism, a documented Escalation Ceiling, not just a stated intention. A Red Line applies throughout: an agent that acts on a person's

body or immediate physical space without consent, outside a narrowly pre-authorized emergency override, scores zero regardless of its other properties.

3.2 Design Principles

Four principles run through all three pillars:

- Feasibility before value: a zero-efficacy score under the Feasibility Floor is a statement about this moment and this place — “not now, not here” — not a judgment on the action's merit in general.
- Permission is plural: legal permission, the person's own adoption, and a caregiver's or guardian's standing authority are three separate gates, not one. Satisfying one does not satisfy the others.
- Bounded autonomy, not binary autonomy: independent action is scored by whether it stays within a documented ceiling and preserves genuine choice, not simply by whether the agent acts on its own at all.
- Consent is a floor, not a factor to weigh: physical or bodily action without consent is disqualifying on its own; it is never offset by a high score elsewhere.

4. Implementation and Operational Logic

C-E-A operationalizes the framework through a fixed evaluation sequence, applied at each decision point:

1. Apply the Feasibility Floor before scoring Efficacy on its 0–3 rubric scale; a failed floor ends the Efficacy evaluation at zero.
2. Check all three Compliance sub-components independently — legal/regulatory, practical/adoptive, and multi-party permission — since any one gap makes the action non-compliant.
3. Verify Autonomy's operational mechanism: confirm a documented escalation ceiling bounds the agent's independent action, and confirm the Red Line has not been crossed.
4. Record the result against the Unit of Analysis and repeat at the next decision point, building a score history for the relationship rather than a single static rating.

Table 1 presents the 0–3 rubric that steps 1–3 apply to each pillar.

Pillar	0	1	2	3
Compliance	Not legal or medically appropriate, and/or no required party has consented.	Meets one required permission (e.g., legal) but not others — permitted on paper only.	Meets most required permissions; one gap remains (e.g.,	Meets all required permissions: legal, physical/medical, and every relevant party's consent.

Pillar	0	1	2	3
			caregiver scope unclear).	
Efficacy	Fails the Feasibility Floor, or passes it with no plausible benefit.	Passes the floor; benefit is theoretical only, not yet realized under real conditions.	Passes the floor; realized benefit in some but not all relevant circumstances.	Passes the floor; realized benefit under this recipient's actual, current conditions.
Autonomy	Cannot act without a human trigger, or acts on the person's body/space without consent outside a pre-authorized emergency override (Red Line).	Acts independently but offers no real choice to the person.	Acts independently with choice offered, but no documented Escalation Ceiling, or support is inconsistent.	Acts independently within a documented Escalation Ceiling; the person or caregiver has a clear, supported choice at every step.

Table 1. C-E-A 0–3 assessment rubric

The 0 row is not symmetric across pillars: for Autonomy it marks two distinct failure modes — an agent too passive to act, and an agent that overrides consent — deliberately scored the same so that neither is read as merely a low, gradable score.

An optional Implementation Mapping extension connects each pillar to a domain's own system architecture and existing trust or safety indices, so C-E-A does not compete with domain vocabulary that already exists. Section 5.2 illustrates this mapping for personalized robotics.

5. Illustrative Case: Personalized Robotic Care

Personalized robotic care for older adults — a household robot offering continuous, single-person physical assistance — exemplifies why C-E-A needs all three pillars evaluated jointly rather than as a single composite score.

Personalized robotic care of this kind — a home robot exercising continuous, autonomous physical judgment — remains largely anticipatory rather than currently deployed at scale; it is used here as a forward-looking case precisely because it forces C-E-A's hardest questions (feasibility under real physical constraints, multi-party consent, bounded autonomy) into concrete form before deployment decisions are made.

5.1 Compliance, Efficacy, and Consent in Embodied Assistance

- No natural event boundary: unlike a one-off evacuation alert, a home robot's relationship with a person has no obvious start or end point, which is exactly why the Unit of Analysis scores individual action-instances rather than the relationship as a whole.

- Zero efficacy is a real outcome, not an edge case: an assistance action that cannot be performed in the person's real space — blocked by clutter, stairs, or a closed door — has zero efficacy no matter how beneficial it would be under ideal conditions.
- Caregivers add a permission layer the person's own consent cannot substitute for: a home robot has a caregiver-set permission profile; an action can be legal, medically appropriate, and welcomed by the person, and still be non-compliant if it falls outside the caregiver's authorized scope.
- The person is never a patient of the robot: this principle underlies the Autonomy Red Line in practice — the robot may act protectively, but the person retains complete spatial and bodily authority outside a narrow, pre-authorized emergency override.

5.2 C-E-A Dimensions in Personalized Robotic Care

Table 2 maps each pillar to its constraints and example application in this case; a full illustrative vignette is provided in Appendix A.

C-E-A Pillar	Core Question	Constraints in Personalized Robotic Care	Example AI-Guided Application
Compliance	Is this action legally and medically permitted, consistent with the person's own physical constraints, and accepted by everyone who holds standing permission over it?	A caregiver-set permission profile sits alongside legal and medical rules; an action can be legal, medically appropriate, and welcomed by the person and still fall outside the caregiver's authorized scope.	The robot checks the caregiver-authorized action list before offering physical assistance, even when the person requests something the robot could otherwise safely do.
Efficacy	Will this action actually prevent harm or protect health, right now, given real conditions?	An action that cannot be performed in the person's real, physical space has zero efficacy regardless of theoretical benefit — there is no partial credit for a good idea that does not fit the room.	The robot checks the home's actual layout and the person's current physical state before proposing an assistance action, not just whether the action would help in principle.
Autonomy	Can the agent act on its own safely, while the person retains real choice and dignity, and does it ever treat the person as a passive object of its action?	A continuous relationship has no natural pause point at which to check that the robot is still respecting the person's agency; the risk is gradual drift toward passive compliance rather than one bad decision.	The robot operates within a documented escalation ceiling and defers to the person or caregiver beyond it; it never acts on the person's body or immediate space without consent outside a pre-authorized emergency.
Unit of Analysis	What, exactly, is being scored — the robot, the	This case's continuous, single-person relationship has	Each offer of assistance, alert, or escalation is scored

C-E-A Pillar	Core Question	Constraints in Personalized Robotic Care	Example AI-Guided Application
	relationship, or the moment?	no obvious event boundary, unlike a one-off alert to a neighborhood.	as its own action-instance; the robot accumulates a running score history rather than one lifetime rating.

Table 2. C-E-A dimensions in personalized robotic care

This case illustrates that no single pillar is sufficient on its own: a legally compliant action that cannot be physically executed provides no benefit, and a highly efficacious action taken without consent is disqualified regardless of its benefit. C-E-A's contribution is treating all three as independently necessary gates rather than folding them into one composite score.

6. Safety, Trust, and Governance

- **Safety:** the Feasibility Floor and the Autonomy Red Line jointly prevent two distinct failure modes — attempting an action the physical environment cannot support, and acting on a person's body without consent.
- **Trust:** a documented Escalation Ceiling is an auditable design artifact, not a stated intention; caregivers and institutions can inspect it directly rather than relying on the agent's self-report of restraint.
- **Governance:** Multi-party Permission Compliance requires an explicit policy on how caregiver, guardian, and institutional authority relate to one another and to the person's own consent — a policy C-E-A deliberately leaves to deployment context rather than fixing in advance.

That last point is a known tension rather than a settled design choice: requiring caregiver permission protects against overreach, but a caregiver's veto can also override a capable person's own wishes. C-E-A does not adjudicate how to weigh a capable person's stated preference against a caregiver's standing permission when the two conflict; it treats this as a governance question for deployment settings to resolve, rather than a question the framework resolves in advance.

7. Implications

- **Chronic disease and home health monitoring:** multi-day, low-drama decisions — medication reminders, mobility support — raise the same caregiver-permission and feasibility questions as acute robotic care, at lower stakes but higher frequency.
- **Population-facing alert systems:** the same three-pillar logic applies to informational agents, not only embodied ones — a digital-only evacuation alert sent to a neighborhood with many low-digital-literacy residents raises the same compliance, efficacy, and autonomy questions as a physical intervention.

- Agentic software beyond robotics: any agent that recommends, schedules, or executes consequential actions on a person's behalf faces the same three questions, whether or not it has a physical body.
- Cross-disciplinary positioning: C-E-A's Autonomy pillar overlaps with, but re-scopes, the autonomy of bioethical principlism²; its Compliance pillar draws on implementation-science constructs⁴ such as feasibility and acceptability without adopting their full measurement apparatus.

8. Conclusions

In contexts where AI agents act rather than merely inform, evaluating a recommended output for accuracy is not sufficient. C-E-A demonstrates that:

- Compliance, Efficacy, and Autonomy are jointly necessary, not substitutable — a deficiency on one pillar cannot be offset by strength on another.
- A hard Feasibility Floor and a hard Autonomy Red Line matter more than fine-grained scoring on the rest of each 0–3 scale: they catch the failure modes with the most immediate consequences.
- Multi-party permission and preserved human agency remain open governance questions — C-E-A surfaces these tensions rather than resolving them, leaving them to deployment-specific policy.

Limitations

C-E-A has been illustrated against a single case — personalized robotic care — and has not yet been tested against non-embodied or population-facing agents, or against the scenarios named below. The framework also does not adjudicate several questions it surfaces: how to weigh a capable person's stated preference against a caregiver's standing permission, whose judgment sets the Efficacy score when experts disagree, and how the Multi-party Permission layer scales beyond a small set of permission-holders. These are treated as open governance and design questions for deployment settings and future work to resolve, rather than as limitations the framework resolves on its own.

Future work will test C-E-A against additional scenarios — including slow-onset emergencies, contested-efficacy situations, agent–institution disagreement, and many-authority settings — to further examine the boundary conditions of each pillar. These open questions are presented here for others working on AI agent evaluation in public health, aging services, and adjacent fields to test and build on.

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Ethics Approval: This is a conceptual/perspective paper using an illustrative, non-identifiable case vignette and does not involve human subjects research or identifiable patient data.

Data Availability: No new data was generated or analyzed in this conceptual paper.

Appendix A: Illustrative Case Vignette — Personalized Robotic Care

An older adult living alone has a home care robot, configured with a permission profile set jointly by the person and their adult child, who serves as caregiver of record. The robot's standing instructions authorize supervised mobility assistance, medication reminders, and fall detection with automatic alert, but require explicit caregiver approval for any new category of physical assistance.

One afternoon, the robot detects that the person has been seated in the same position for an unusually long time and appears mildly unsteady on a prior attempt to rise. A prediction module flags a moderate fall risk if the person attempts to walk to the kitchen alone.

- Compliance: assisting with a supervised walk to the kitchen falls within the robot's caregiver-authorized scope. Physically bracing or lifting the person, by contrast, would exceed the authorized category and require a new caregiver approval before the robot may act, even though it might be medically reasonable and welcomed by the person.
- Efficacy: the robot checks the actual path to the kitchen — a rug has bunched near the doorway. Offering supervised walking assistance along that specific path has zero efficacy until the hazard is addressed; the robot instead proposes clearing the rug first, then re-offers the walking assistance.
- Autonomy: the robot offers, rather than insists: it proposes the supervised walk and waits for the person's response, consistent with its documented escalation ceiling for mobility support. If the person declines, the robot logs the moment and does not repeat the offer

more frequently than its ceiling allows, escalating to the caregiver only if the person shows signs of distress or a subsequent fall risk indicator.

- Red Line in practice: if the person began to fall while the robot was nearby, a narrowly pre-authorized emergency override permits the robot to physically intervene to prevent injury — the one circumstance in which acting on the person's body without a fresh, in-the-moment consent is permitted, because it was authorized in advance for exactly this scenario.

This vignette illustrates why C-E-A treats Compliance, Efficacy, and Autonomy as three separate checks rather than one score: the robot's action is efficacious only once the physical hazard is addressed, compliant only within the caregiver's pre-authorized categories, and autonomous only insofar as the person retains a genuine, supported choice at each offer — with the Red Line reserved for the single, narrowly defined emergency case where all three questions collapse into an immediate safety override.